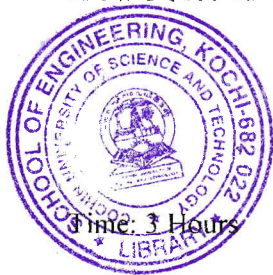


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B.Tech. Degree VIII Semester Supplementary Examination
August 2018



ME 801 OPERATIONS MANAGEMENT
(2006 Scheme)

Maximum Marks: 100

PART A
(Answer ALL questions)

(8 × 5 = 40)

- I. (a) Define the terms critical path, dummy activity and crashing in project network analysis.
- (b) Write note on moving average method of forecasting.
- (c) With sketch explain product life cycle of a product.
- (d) Explain Bin cards.
- (e) Describe GANTT chart with example.
- (f) Explain reaction rate method for Aggregate planning.
- (g) Describe systematic layout planning.
- (h) Compare Preventive and Breakdown maintenance.

PART B

(4 × 15 = 60)

- II. A small project is composed of the following activities whose time estimates in days are given below. Draw the project network, calculate project duration. What is the approximate probability that the jobs on the critical path will be completed in (i) 41 days? (ii) 35 days? (15)

Activity	Optimistic time (days)	Most likely time (days)	Pessimistic time (days)
1 – 2	3	6	15
1 – 6	2	5	14
2 – 3	6	12	30
2 – 4	2	5	8
3 – 5	5	11	17
4 – 5	3	6	15
6 – 7	3	9	27
5 – 8	1	4	7
7 – 8	4	19	28

OR

- III. Explain in detail the classification of short range, medium range and long range forecasting methods. (15)

(P.T.O.)

- IV. (a) Briefly explain the structure of inventory problems. Also write note on relevant cost. (8)
- (b) The rate of use of a particular raw material from stores is 20 units per year. The cost of placing and receiving an order is ₹40. The cost of carrying inventory is 16% per year. Determine the economic order quantity. (7)

OR

- V. Write notes with sketches on,
- (a) Material requisition sheet. (5)
- (b) Material transfer note. (5)
- (c) Material return note. (5)

- VI. Seven jobs are to be processed through 3 machines M1, M2 and M3 in the order M1, M2, M3. The processing times are given in hours to process each one of the 3 jobs through all the machines. Find the optimal sequence of the jobs. Also find the total elapsed times and idle times on M2 and M3. (15)

Jobs	M1	M2	M3
A	3	4	6
B	8	3	7
C	7	2	5
D	4	5	11
E	9	1	5
F	8	4	6
G	7	3	12

OR

- VII. Sequence the jobs waiting to be processed in a machine shop using critical ratio method for the following data. Assume today is the first day in the calendar. (15)

Job	1	2	3	4	5
Processing time (days)	12	4	6	8	10
Due date on Calendar	36	12	18	22	16

- VIII. (a) What are the principles of material handling? Explain. (8)
- (b) Explain in detail different types of maintenance procedures. (7)

OR

- IX. (a) Explain with example salient features of different plant layouts with suitable sketches. (10)
- (b) What are the reasons for replacement of machines in industries? (5)
